MIL-B-18907A(OS)
29 April 1988
SUPERSEDING
MIL-B-18907(NOrd)
30 June 1955

MILITARY SPECIFICATION

BANDS, PROJECTILE ROTATING

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification augments pertinent drawings and specifications shown on the list of drawings and specifications particular to the item of the contract so that a study of the contract, drawings, and specifications will clearly set forth the manufacturing and inspection requirements for projectile rotating bands which, before assembly, are blanks in the form of rings of drawn, forged, or rolled material.

1.2 Classification. Projectile rotating bands shall be of the following materials, as specified in Table I.

Copper
Gilding Metal
Cupro-Nickel

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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FEDERAL

FED-STD-151	Metals; Test Methods
DOD-STD-2101	Classification of Characteristics

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-8	Metallic Materials, Tension Testing of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be as specified on the applicable projectile drawing.

3.1.1 Chemical composition and physical requirements. Chemical composition and physical requirements shall be as specified in tables I and II.

3.2 Assembly.

3.2.1 Annealed blanks. Projectile rotating band blanks shall be in the annealed condition when assembled.

3.2.2 Cold blanks. Blanks for projectiles smaller than 6 inches shall be assembled cold.

3.2.3 Seating. Band seat shall be free of all foreign material prior to assembly of the band. Blanks shall be pressed to a tight fit in the roughened band seat in such a manner as not to deform the projectile body. The assembled band shall completely fill the band seat. Bands shall withstand the seating test specified in 4.3.1.

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TABLE I. Chemical composition (see 4.4).

Material	Copper[1] (percent)	Iron (maximum percent)	Lead (maximum percent)	Other elements (percent)	Total impurities (maximum percent)
Copper	99.88[2]	—	—	—	0.12
Gilding metal	89-93	0.05	0.05	Zinc[3]	0.125
Cupro-Nickel	—[3]	0.10	—	Nickel 2.0-3.0 Manganese 0.40[5]	0.30[4]

[1] Any silver present shall be counted as copper.
[2] Minimum.
[3] Remainder.
[4] Elements not otherwise specified.
[5] Maximum.

TABLE II. Physical properties (from tensile tests, see 4.3.2).

Material	Caliber of projectile (inches)	Tensile strength (minimum, lb/in^2)	Elongation (minimum percent)		For test specimen, see	Condition
			in 1 inch	in 2 inches		
Copper	Less than 1.0	30,000	30	—	4.3.2	—[1]
	1.0 to 1.875	30,000	30	—	4.3.2.2	—[1]
	1.88 to 4.50	30,000	—	15	4.3.2.2	—[2]
	4.51 to 5.50	30,000	—	25	4.3.2.2	—[2]
	5.51 to 7.50	30,000	—	30	4.3.2.2	—[2]
	Over 7.50	30,000	—	30	4.3.2.2	—[2]
Gilding metal	Less than 1.0	32,000	30	—	4.3.2	—[1]
	1.0 to 1.875	32,000	30	—	4.3.2.2	—[1]
	1.88 to 4.50	35,000	—	15	4.3.2.2	—[2]
	4.51 to 5.50	35,000	—	25	4.3.2.2	—[2]
	Over 5.50	35,000	—	30	4.3.2.2	—[2]
Cupro-Nickel	Over 7.50	30,000	—	30	4.3.2.2	—[2]

[1] Test specimens taken from annealed blanks before assembly.
[2] Test specimens taken from bands removed from projectiles after assembly but before machining.

3.2.4 Heated blanks. Blanks for 6-inch and larger projectiles may be heated and assembled hot, provided the blank is not heated to a temperature above 1650° F and provided the furnace is fitted with pyrometric control to insure uniformity of assembly. Blanks assembled hot shall be allowed to cool to at least 750° F under the last application of banding pressure. Care shall be taken to prevent oxidation during heating. Blanks which are heated for hot assembly to a temperature higher than the annealing temperature need not be annealed before being heated for assembly.

3.3 Bending. Bands shall withstand the bending test specified in 4.3.3 without cracking on the outside of the bent portion.

3.4 Workmanship. All blanks shall be free from slivers, pipes, laps, cracks, laminations, grooves, seams, oxidation flaws, deep scratches, and other defects which may render them unsuitable for the purpose intended. All blanks shall be round, of uniform dimensions, and free from scale, chips, oil, grease, and dirt.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105 (see 6.2.1).

4.2.1 Inspection lot. As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2.2 Production lot. Unless otherwise specified each production lot of assembled band shall be the same quantity as required by the applicable projectile specification.

4.2.3 Sampling for destructive tests. At least one band shall be carefully removed from each production lot, before machining, to permit examination of seating and physical tests. A minimum of two samples shall be taken for physical testing. Multiple test samples may be taken from the same band. During production, if a rejected lot occurs, retest may be performed on an additional 30 samples from the rejected lot. The destructive test shall be performed in such a way that a minimum number of rotating bands are required (i.e., multiple test samples may be taken from the same rotating band). Acceptance criteria shall be accept on zero, reject on one.

4.3 Test procedures.

4.3.1 Seating. The seating of assembled bands shall be tested before machining by tapping the band lightly with a steel hammer. A hollow sound accompanied by an unusual rebound of the hammer indicates an improperly seated band. A more critical test may be used for the examination of doubtful cases such as laying a small coin (a dime of U.S. currency) on the band and tapping the band near the coin with a 4-ounce hammer. An unseated condition under the coin will be shown by a marked rebound of the coin.

4.3.1.1 At least one band, and a greater number as the Inspector considers necessary, shall be carefully removed from each lot of projectiles to permit examination of the seating. If there is an indication that the assembly has not been performed satisfactorily, all projectiles in the lot shall be rebanded. Projectiles with improperly seated bands and those from which bands have been removed may be rebanded and resubmitted for test.

4.3.2 Tension tests. Tension testing shall be in accordance with ASTM E-8.

4.3.2.1 Small bands. For blanks with an inside diameter less than 1 inch, a sample of the tubes from which the blanks are cut shall have the full section tested in accordance with ASTM E-8. If blanks are cut from hard tubing and then annealed, the test section shall be annealed with the blanks which it represents.

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4.3.2.2 Large bands. For blanks or bands with an inside diameter of 1 inch or more, tension test specimens shall be cut in the condition indicated in table II. If length is sufficient, as determined by ASTM E-8 Figure 13, a longitudinally cut band specimen shall be used (e.g. 16-inch projectiles); otherwise specimens may be circumferentially cut. Circumferentially cut specimens shall be straightened cold (at ambient temperature) by pressure, without hammering, and otherwise prepared in a manner as to minimize cold-working.

4.3.2.3 Specimen thickness. The thickness of test specimens shall be the full thickness of the material, except that a small amount may be removed by machining, when necessary, to obtain a suitable test specimen.

4.3.3 Bending test. For all sizes, the bending test shall be made on specimens taken from bands removed from projectiles after assembly but before machining. Jaw marks and similar surface imperfections may be removed by machining and sharp corners may be "broken" with a smooth file.

4.3.3.1 Small bands. In case of small bands, bending test specimens shall consist of whole bands cut open on one side.

4.3.3.2 Large bands. In case of large bands, bending test specimens shall consist of a section of the full width and thickness, with length approximately one-third the circumference.

4.3.3.3 Procedure. The specimen shall be bent cold by pressure, without hammering, in the direction of curvature of the blank, until it tightly encloses 180° of a rod equal in diameter to the thickness of the specimen.

4.4 Chemical composition. Chemical analysis of the band material shall be determined by testing suitable specimens removed from each lot of raw material. Testing shall be in accordance with FED-STD-151, Method 111.2 or 112.2. A certified analysis from the material producer is required; however, it shall not relieve the contractor from performance of this test.

4.5 Classification of characteristics. In accordance with DOD-STD-2101 the following classifications apply:

M101 Chemical Requirements, table I
M102 Physical Properties, table II
M103 Band Seating, paragraph 3.2.3
M104 Bending Requirements, paragraph 3.3

5. PACKAGING

5.1 Packaging and packing. Projectile rotating bands shall be packaged and packed in accordance with the contract requirements (see 6.2.1).

6. NOTES

6.1 Intended use. The projectile rotating band around the base of a projectile centers the projectile and makes it fit tightly in the bore, thus preventing the escape of gas and by engaging the rifling, gives the projectile its spin.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type of material (see 1.2)
c. Sampling plan, if other than as specified in 4.1
d. Quantity of assembled bands per production lot (see 4.1.2)
e. Packaging and packing requirements (see 5.1)

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6.3 Subject term (key word) listing.

Bands
Projectiles
Projectile rotating bands

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - OS
Project 1320-N B13

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-18907A(OS)	BANDS, PROJECTILE ROTATING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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